Lab 5: COVARIANCE MATRIX AND CORRELATION MATRIX

DUE: 13:00 22/05/2026

Submission: mlearning

- **Lab#_YOURNAME.ipynb**
- Must include all data file (**.csv,…**) and **.ipynb** in your submission
- **.csv** and **.ipynb** must be in the same folder (pd.read_csv('data.csv'))
- No use of LOCAL libraries/folders
- NO LATE submission accepted

Use '**house_prices.csv**'.

The selected features are highlighted:

**Id MSSubClass MSZoning LotFrontage LotArea Street Alley LotShape
LandContour Utilities LotConfig LandSlope Neighborhood Condition1
Condition2 BldgType HouseStyle OverallQual OverallCond YearBuilt
YearRemodAdd RoofStyle RoofMatl Exterior1st Exterior2nd
MasVnrType MasVnrArea ExterQual ExterCond Foundation BsmtQual
BsmtCond BsmtExposure BsmtFinType1 BsmtFinSF1 BsmtFinType2 BsmtFinSF2
BsmtUnfSF TotalBsmtSF Heating HeatingQC CentralAir Electrical
1stFlrSF 2ndFlrSF LowQualFinSF GrLivArea BsmtFullBath BsmtHalfBath
FullBath HalfBath BedroomAbvGr KitchenAbvGr KitchenQual
TotRmsAbvGrd Functional Fireplaces FireplaceQu GarageType
GarageYrBlt GarageFinish GarageCars GarageArea GarageQual GarageCond
PavedDrive WoodDeckSF OpenPorchSF EnclosedPorch 3SsnPorch
ScreenPorch PoolArea PoolQC Fence MiscFeature MiscVal
MoSold YrSold SaleType SaleCondition SalePrice**

a) Computer the covariance matrix and plot a heatmap of the covariance matrix of selected features using **Seaborn** package
b) Plot the correlation heatmap of the of the correlation matrix of selected features using **Seaborn** package
c) Which features are highly correlated with SalePrice? Among these features, how many are highly correlated with each other? Select a few features for SalePrice prediction based on your observations. Explain your choice.

d) Plot scatterplots of 5 highly-corelated pairs of features and 5 non corelated pairs of features.

e) Use **pandas.plotting.scatter_matrix** for the selected features.

Note:

pandas.plotting.scatter_matrix is a function within the pandas library that generates a matrix of scatter plots and histograms (or kernel density estimates) for a given DataFrame. This visualization tool is particularly useful for exploring relationships and distributions among multiple numerical variables in a dataset.

- **Scatter Plots:**
  For every unique pair of numerical columns in the input DataFrame, scatter_matrix creates a scatter plot, allowing for the visual inspection of bivariate relationships and potential correlations.

- **Histograms/KDEs:**
  Along the diagonal of the matrix, the function displays either histograms or Kernel Density Estimation (KDE) plots for each individual numerical variable. These plots illustrate the distribution of each variable.